

How Reconstruction Errors Affect the FlyWire Connectome

Measuring how wiring diagram errors change network structure across 5 fruit fly connectomes

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WHY THIS MATTERS

Automated AI tools make mistakes when mapping brain wiring diagrams from electron microscope images. This benchmark tests how much these errors change network connections, brain components, and neuron importance rankings.

DATASETS TESTED

- 5 Drosophila Connectomes: BANC (158k), FAFB (139k), MANC (24k), MCNS (167k), MAOL (52k neurons)
- 1,030 independent simulation runs across 10 error levels (0% to 20%)
- Tracked network connections, connected components, and PageRank rankings

KEY PAPERS & REFERENCES

- Dorkenwald et al. Nature 2024 (FlyWire Whole-Brain Connectome)
- Buhmann et al. Nature Methods 2021 (Automated Synapse Detection)
- Januszewski et al. Nature Methods 2018 (Flood-Filling Segmentation)
- Takemura et al. Nature 2023 | Scheffer et al. eLife 2020

THE FIVE ERROR TYPES TESTED

1. Missed Synapses (EM1)

The detector fails to spot a real synaptic connection.

2. False Synapses (EM2)

The detector adds a fake connection where none exists.

3. Synapse Count Jitter (EM3)

Small counting errors in individual synapse numbers.

4. Split Neurons (EM4)

A single biological neuron gets broken into fragments.

5. Merged Neurons (EM5)

Two distinct neurons get accidentally joined together.

MAIN GOAL

Provide clear sensitivity benchmarks showing how each error type affects brain graph measurements.

How Each Error Type Changes the Graph

Comparing cross-dataset averages and individual connectome ranges

OVERALL ERROR IMPACT (AT 20% ERROR)

Each error type produces a distinct pattern of change across the connectome.

RESULTS AT HIGHEST TESTED ERROR LEVEL (20%)

Error Model	Δ Edge Count	Δ Synapses	Δ Var
Missed Synapses (n=5)	-4.9%	-20.0%	-30.3%
False Synapses (n=3)	+19.4%	+7.6%	-13.8%
Synapse Count Noise (n=5)	0.0%	+0.03%	+5.5%
Split Neurons (n=5)	0.0%	0.0%	0.0%
Merged Neurons (n=4)	-10.9%	-0.1%	+46.9%

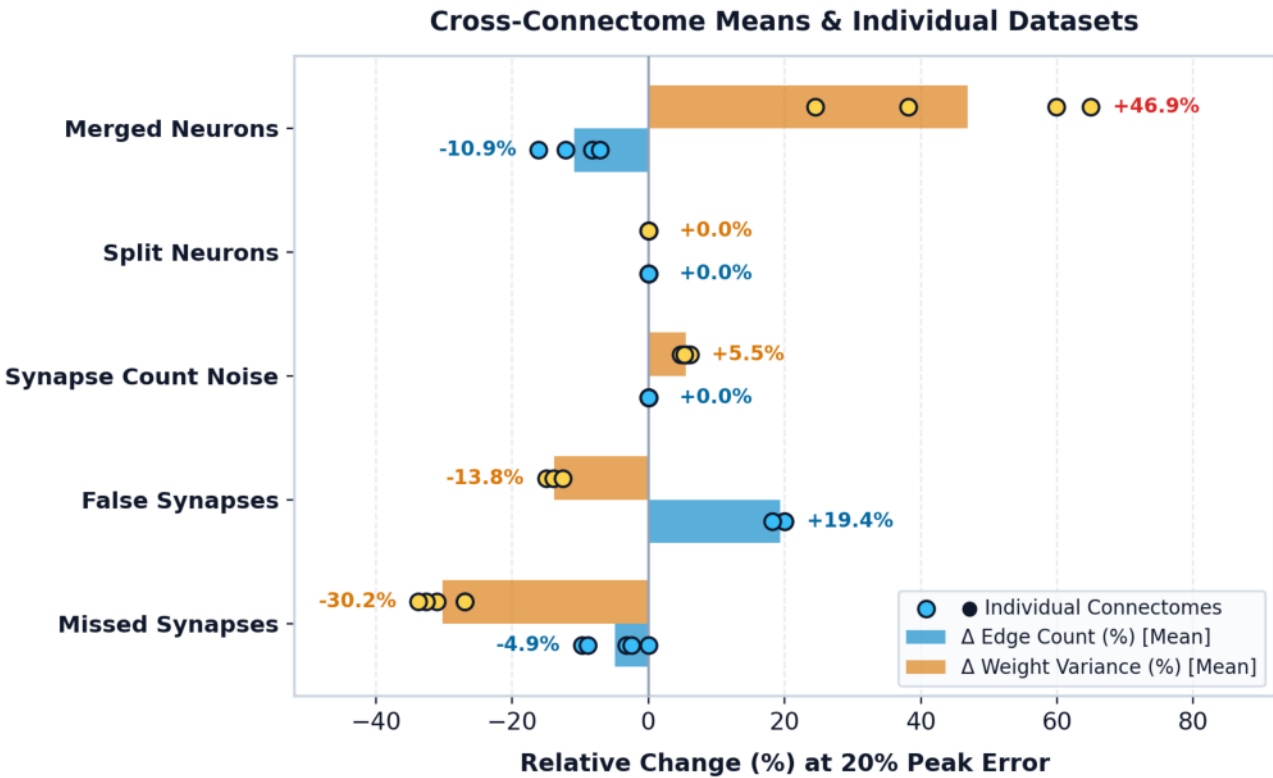
INDIVIDUAL CONNECTOME RANGES

- Missed Synapses: MCNS (-0.007%) to MANC (-9.7%)
- Merged Neurons: MCNS (-7.1%) to FAFB (-16.1%)
- Splits & Noise: 0.0% edge loss across all 5 datasets

MAIN TAKEAWAY

Merged neurons caused the largest structural damage (-10.9% mean edge loss, +46.9% variance surge). Splits and noise preserve total connections.

Note: n varies by model (n=3 to 5). At typical error levels (2-5%), metrics shift by <1%.



Why Missed Synapses Do Not Proportionally Eliminate Connections

Connections with multiple synapses provide built-in protection against loss

SYNAPSE LOSS VS CONNECTION LOSS

Evaluating whether losing 20% of synapses removes 20% of graph connections.

MAIN RESULTS

- Total synapses drop by ~20.0%
- Connection count drops by only -4.87% (~4x smaller than synapse loss)
- Largest Connected Core (SCC) shrinks by only -0.04%

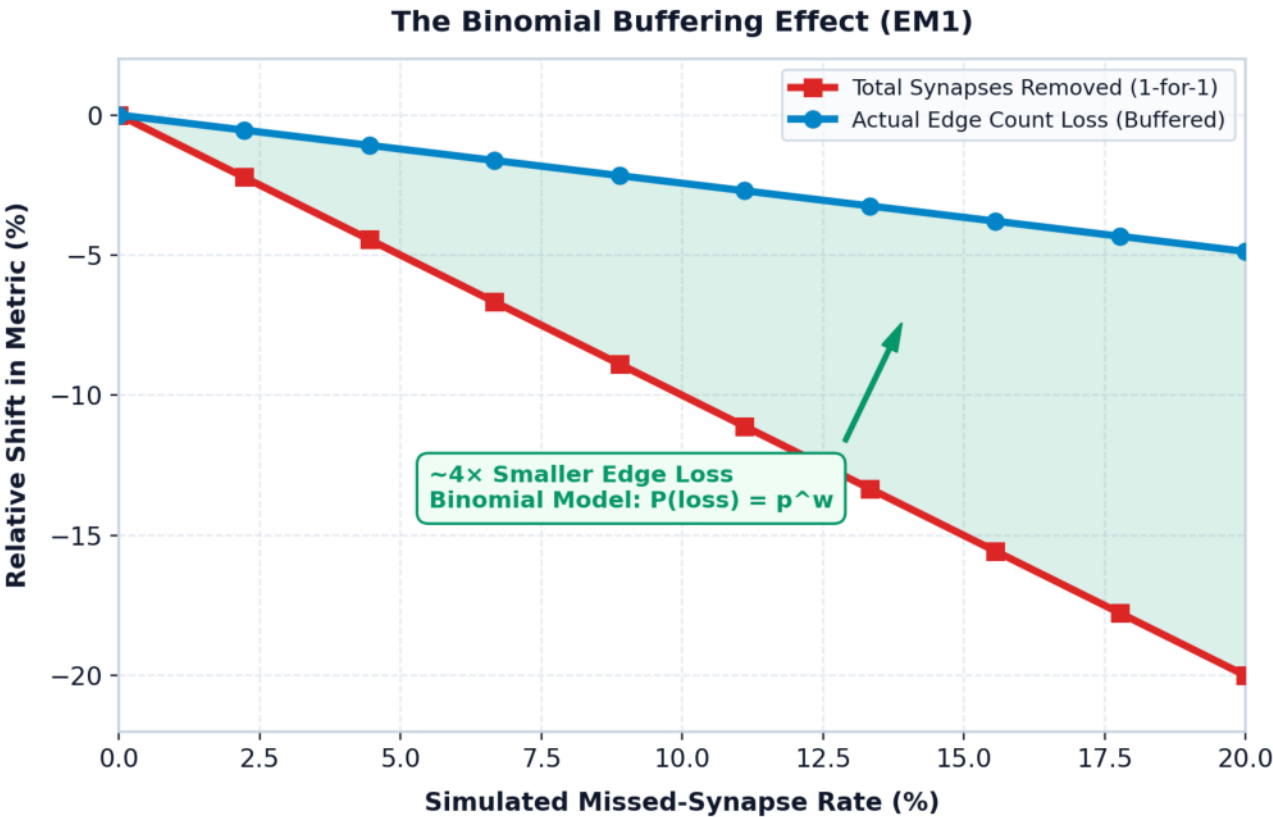
WHY THIS HAPPENS

In real connectomes, many connections have multiple synapses. A connection is only lost if every single synapse on it is missed ($P = p^w$). Under a 20% miss rate (illustrative example):

- 1-synapse edge: 20.0% chance of loss
- 5-synapse edge: 0.03% chance of loss

Connections with more synapses are strongly protected from disappearing.

Error model rates are motivated by automated synapse detection studies (Buhmann et al. 2021).



Merging Neurons Produces Larger Changes in Edge Count and Weight Variance

Neuron merges reduce edge count and strongly increase connection-weight variance

SPLITS VS MERGES

Comparing what happens when neurons are split into pieces versus when separate neurons are accidentally merged.

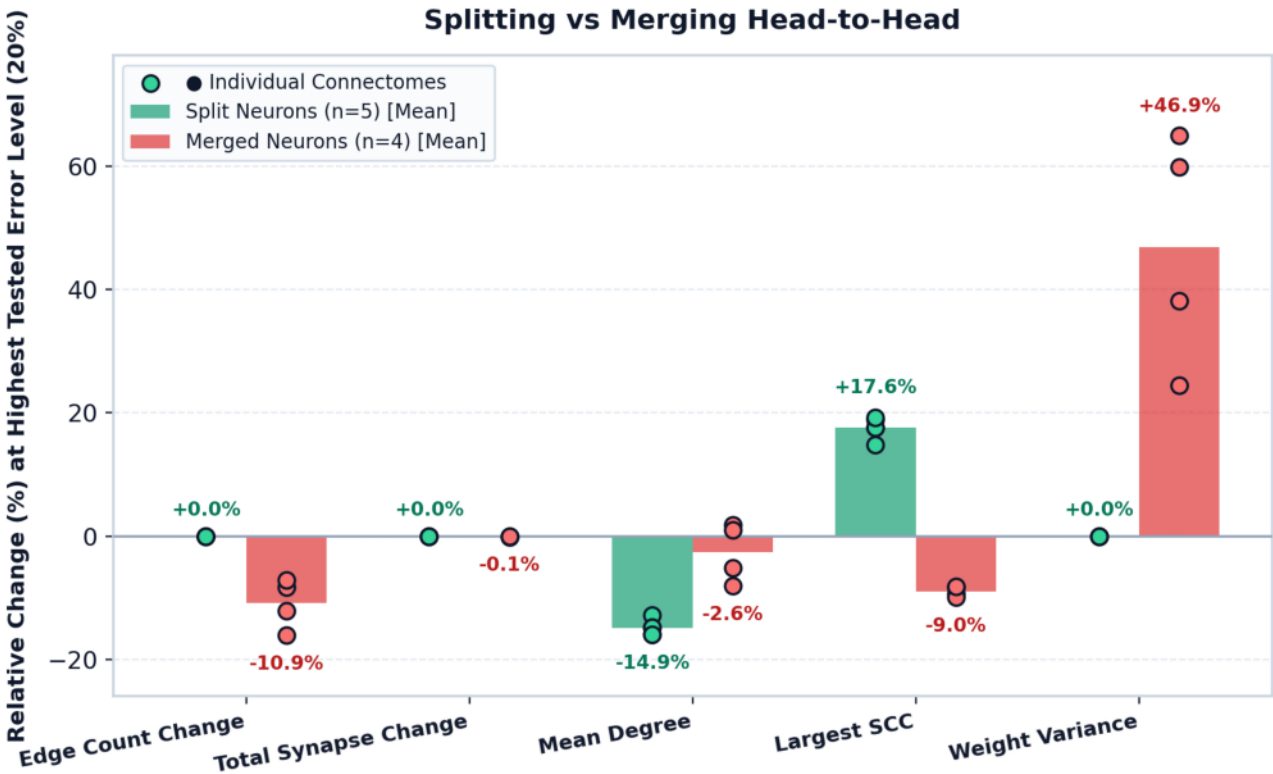
HEAD-TO-HEAD COMPARISON (AT 20% ERROR)

Metric	Split Neurons (n=5)	Merged Neurons (n=4)
Edge Count Change	0.0%	-10.9%
Total Synapse Change	0.0%	-0.1%
Mean Degree	-14.9%	-2.6%
Largest Strongly Connected Component (SCC)	+17.6%	-9.0%
Weight Variance	0.0%	+46.9%

KEY TAKEAWAY

Splitting redistributes existing connections across fragments while preserving total edge and synapse counts. Merging combines neurons, reducing edge count by 10.9% and increasing weight variance by 46.9%.

Splits and merges are important reconstruction error types in automated EM segmentation (Dorkenwald et al., 2024; Januszewski et al., 2018).



PageRank Remains Stable; Higher Connection Strength Is Associated With Lower Edge Loss

PageRank similarity to baseline remains high ($r \geq 0.977$); edge loss varies with median connection weight

1. PAGERANK SIMILARITY TO BASELINE REMAINS HIGH

- PageRank remains highly correlated with the unperturbed baseline across the tested error models.
- Pearson correlations range from approximately 0.977 to 1.000 (mean $r = 0.999$ under missed synapses).
- PageRank rankings remain highly similar to the baseline under the tested perturbations.

2. HIGHER CONNECTION STRENGTH IS ASSOCIATED WITH LOWER EDGE LOSS

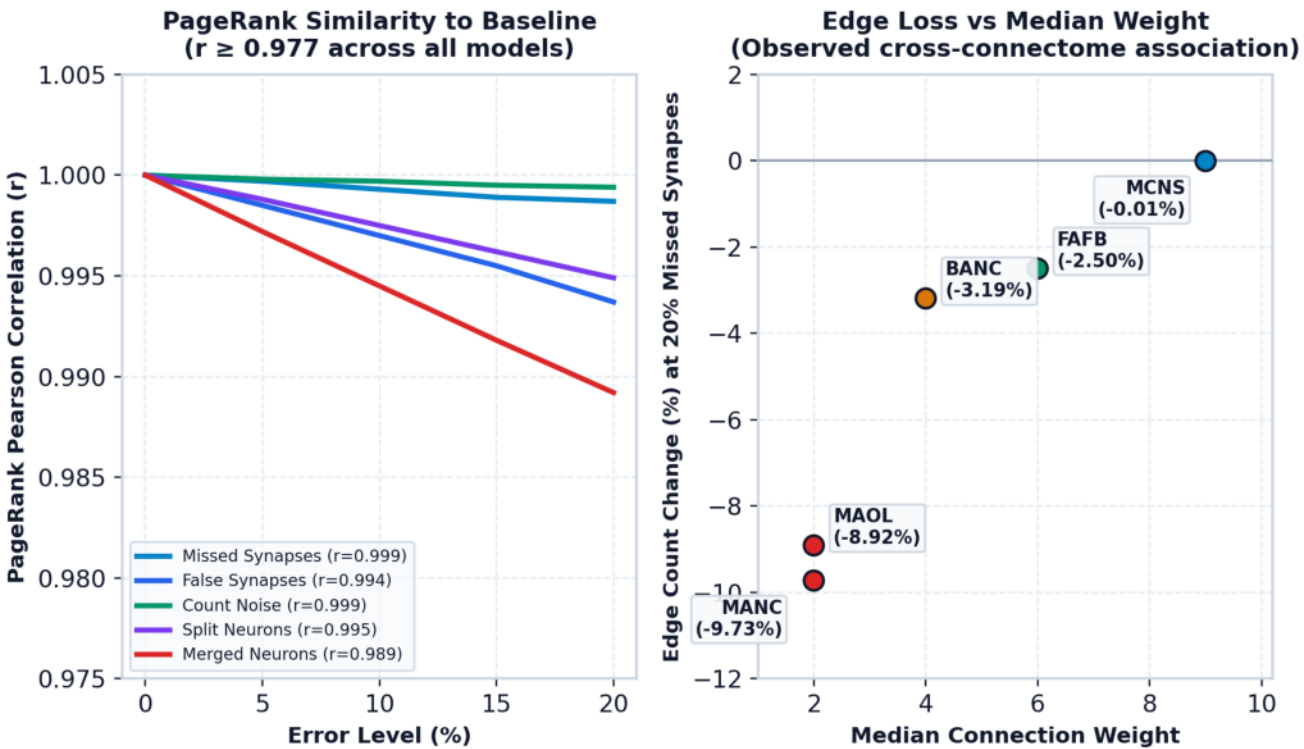
Under the 20% missed-synapse perturbation, observed edge loss varies across connectomes with different median connection weights:

- MCNS (median weight 9.0): -0.007% edge count change
- FAFB (median weight 6.0): -2.50% edge count change
- BANC (median weight 4.0): -3.19% edge count change
- MAOL (median weight 2.0): -8.92% edge count change
- MANC (median weight 2.0): -9.73% edge count change

MAIN TAKEAWAY

PageRank remains highly similar to baseline under the tested perturbations, while connectomes with higher median connection weights show lower observed edge loss under the missed-synapse model.

Note: Observations reflect cross-connectome associations across the 5 tested datasets (Scheffer et al. eLife 2020).



Secondary Effects of Reconstruction Errors

Different reconstruction errors leave different fingerprints beyond the quantity directly perturbed.

1. MISSED SYNAPSES

Direct: 20% of synapses are removed

SECONDARY EFFECT:

Many individual synapses can be lost while most neuron-to-neuron connections remain.

Evidence: -20.0% synapses | -4.9% connections | PageRank r = 0.999

2. FALSE SYNAPSES

Direct: New false connections are added

SECONDARY EFFECT:

Many new connections are added, but they are relatively weak, so connection count grows much faster than total synaptic strength.

Evidence: +19.4% connections | +7.6% synapses | -13.8% weight variance

3. SYNAPSE-COUNT NOISE

Direct: Connection weights changed; graph unchanged

SECONDARY EFFECT:

The network structure stays the same, but connection strengths become less reliable.

Evidence: 0.0% connection change | 0.0% degree/SCC | +5.5% weight variance

4. SPLIT ERRORS

Direct: Neurons split into multiple neuron pieces

SECONDARY EFFECT:

Connections are kept, but they are spread across more neurons. This lowers connections per neuron and changes the largest connected group.

Evidence: +17.5% neurons | 0.0% connection loss | -14.9% mean degree | +17.6% largest SCC

5. MERGE ERRORS

Direct: Multiple neurons combined into one

SECONDARY EFFECT:

Most synapses remain, but merging neurons combines separate connections into fewer and much stronger-looking connections.

Evidence: -0.1% synapses | -10.9% connections | +46.9% weight variance | -9.0% largest SCC

KEY QUALITY-CONTROL TAKEAWAY

Each error changes the network in more ways than the direct error itself. Some errors mainly change connections, some change connection strength, and others change how connections are distributed across neurons.

Reliable connectome quality checks therefore need to look at several network measurements, not just one.

BENCHMARK: 5 connectomes • 10 error levels • 1,030 runs | Surjit Mandal

Individual Connectome Breakdown Table

Comparing exact Δ Edge Count and metrics across BANC, FAFB, MANC, MCNS, and MAOL (at 20% Peak Error)

1. MISSED SYNAPSES (EM1) BREAKDOWN

Connectome	Median W	Δ Edge Count	PageRank r
BANC	4.0	-3.19%	0.998
FAFB	6.0	-2.50%	1.000
MANC	2.0	-9.73%	0.998
MCNS	9.0	-0.007%	1.000
MAOL	2.0	-8.92%	0.999
MEAN (n=5)	4.6	-4.87%	0.999

2. MERGED NEURONS (EM5) BREAKDOWN

Connectome	Δ Edge Count	Δ Weight Var	Largest SCC
BANC	-12.18%	+59.87%	-8.30%
FAFB	-16.09%	+64.98%	-9.33%
MANC	-8.27%	+24.46%	-9.98%
MCNS	-7.12%	+38.20%	-8.24%
MEAN (n=4)	-10.91%	+46.88%	-8.96%

TAKEAWAY ACROSS CONNECTOMES

All individual connectomes exhibit the same core phenomenon: edge buffering under synapse loss, severe edge loss under merges, and 100% edge preservation under splits.

3. FALSE SYNAPSES (EM2) BREAKDOWN

Connectome	Δ Edge Count	Δ Synapses	Δ Weight Var
BANC	+20.00%	+10.02%	-15.03%
FAFB	+20.00%	+6.58%	-13.90%
MCNS	+18.18%	+6.32%	-12.61%
MEAN (n=3)	+19.39%	+7.64%	-13.85%

4. SPLIT NEURONS (EM4) ACROSS ALL 5 DATASETS

- Across BANC, FAFB, MANC, MCNS, and MAOL:
- Δ Edge Count: 0.0% (100% preserved in all 5 datasets)
 - Δ Total Synapses: 0.0% (100% preserved in all 5 datasets)
 - Mean Degree: drops by -12.8% to -16.0% (mean -14.9%)
 - Largest Connected Core: shifts by +14.9% to +19.2% (mean +17.6%)

5. BASELINE DATASET CHARACTERISTICS

- BANC: 158k nodes | 3.99M edges | 23.6M synapses (med wt: 4.0)
- FAFB: 139k nodes | 5.34M edges | 50.7M synapses (med wt: 6.0)
- MANC: 24k nodes | 6.24M edges | 30.9M synapses (med wt: 2.0)
- MCNS: 167k nodes | 6.24M edges | 89.8M synapses (med wt: 9.0)
- MAOL: 52k nodes | 6.74M edges | 26.5M synapses (med wt: 2.0)